

Introduction to Tokens and Tokenization

Spoken Tutorial

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<https://spoken-tutorial.org>

Learning Objectives

In this tutorial, we will learn,

- ▶ **Explain what tokens are, how tokenization works, and why tokens are crucial for LLM performance, cost, and efficiency.**



System Requirements

- ▶ **Here I am using a browser on a computer or mobile.**
- ▶ **We will also use an online tokenizer tool.**



Pre-requisite Slide

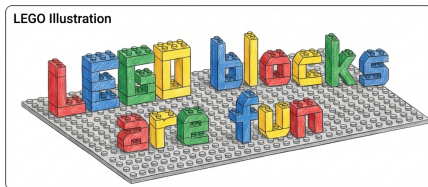
- ▶ **Introduction to Generative AI,
Introduction to Prompting**



What are Tokens?

- ▶ LLMs don't read words like humans do.
- ▶ They process text in small chunks called tokens.

Think of tokens
as the



How Tokenization Works

- ▶ The tokenizer acts like a special dictionary for the LLM.
- ▶ It knows how to split raw text into these sub-word



FROM TEXT TO TOKENS: A VISUAL ANALOGY

TEXT PROCESSING
(CHOPPING)



TOKENIZATION
(SEGMENTING)



Why Tokens Matter: Context Windows

- ▶ Now, why should you care about tokens?
- ▶ First, consider the LLM's context window.



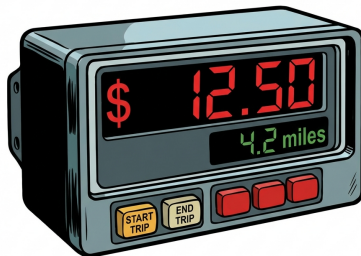
**Think of the
context window**



Why Tokens Matter: Cost

- ▶ Second, tokens directly impact the cost of using LLMs.
- ▶ API providers charge you per token processed.

It's like a taxi



Why Tokens Matter: Chunking

- ▶ Finally, tokens are vital for chunking your documents.
- ▶ When preparing large documents for an LLM, we often break them into



Summary

In this tutorial, we learned that:

- ▶ **With this, we come to the end of this tutorial.**
- ▶ **Tokenization is the process of splitting text into these sub-word units.**
- ▶ **And tokens are critical for context windows, cost, and efficient chunking.**



Assignment

- ▶ Find an online tokenizer tool.
- ▶ Experiment with different types of text: short words, long words, sentences,...
- ▶ Observe how the token count changes.
- ▶ Try to find a sentence that results in exactly 10 tokens.



Thank you

